

Desktop PC Hardware Troubleshooting

Assignment Answers

1. Five common desktop PC hardware issues and symptoms

Hardware issue	Typical symptom
RAM (memory) failure or loose RAM	PC may fail to POST, beep repeatedly, restart, or show a blank screen.
Hard drive/SSD failure	"No boot device" message, slow loading, clicking from an HDD, or files failing to open.
Power supply (PSU) fault	PC will not turn on, switches off unexpectedly, or restarts under load.
Graphics card/display connection issue	Monitor shows no signal, a blank screen, artifacts, or distorted graphics.
CPU overheating or cooler failure	PC becomes unusually hot, fans run loudly, performance drops, or the system shuts down.

2. Laptop is not charging: basic step-by-step troubleshooting

- **Step 1 — Check the wall outlet.** Plug in a lamp or another known-working device, or try a different outlet. This checks whether the outlet is supplying power.
- **Step 2 — Inspect the charger and cable.** Look for cuts, bent pins, fraying, burn marks, or a loose adapter connection. Do not use a damaged charger.
- **Step 3 — Check the power adapter indicator.** If the adapter has an LED, see whether it lights when plugged into the wall and whether it changes when connected to the laptop.
- **Step 4 — Check the laptop charging port.** With everything unplugged, look for dust, damage, or looseness. Do not insert metal objects into the port.
- **Step 5 — Reseat all connections.** Disconnect the charger from the wall and laptop, then reconnect it firmly. If the laptop uses USB-C charging, use a compatible port and a charger with the required power rating.
- **Step 6 — Try a known-compatible charger.** If available, test with a manufacturer-approved or correctly rated charger. If it works, the original charger or cable may be faulty.
- **Step 7 — Perform a basic power reset.** Unplug the charger; if the battery is removable, remove it. Hold the power button for about 15–20 seconds, reconnect power, and try again. Follow the laptop maker's instructions if they differ.
- **Step 8 — Check the result.** Look for the charging light and whether the battery percentage begins to rise. If it still will not charge, the battery, DC-in jack, charging circuit, or motherboard may need professional service.

3. Test a USB phone charger with a multimeter

Safe method: Use a USB breakout/test adapter that exposes the 5V and ground contacts. Set the multimeter to DC volts (a range above 5V, if not auto-ranging). Connect the black probe to ground and the red probe to the +5V contact. Do not let the probes touch each other, and do not push bare probes into a live

USB port.

Example recorded reading (illustrative): 5.08 V DC. This is within the expected 4.75–5.25 V range.

Replace this example with the actual reading if you perform the measurement; no physical measurement was performed for this worksheet.

4. Testing the ATX PSU 12V rail for random shutdowns

Important safety note: A power supply can contain hazardous voltages. Do not open the PSU casing. For this test, disconnect AC power and all PSU connections from the PC, and use a suitable ATX breakout/test adapter if possible. Never probe a live connector by guessing pin positions. If you are not trained to work safely around PC power connectors, ask a technician.

1. Turn off the PC, switch off and unplug the PSU from the wall, and wait briefly for stored charge to dissipate.
2. Disconnect the ATX power connector from the motherboard and other components. Confirm the connector is isolated from the PC.
3. Set the multimeter to DC voltage, on a range that can read 12 V (for example, 20 V DC if the meter is not auto-ranging).
4. Using a suitable breakout adapter, identify the yellow +12 V wire and a black ground wire. Connect the black probe to black/ground and the red probe to yellow/+12 V.
5. For a meaningful powered reading, the PSU must be energized using an appropriate ATX tester or safe test setup, with the connector disconnected from PC components. Do not improvise by shorting pins; if unsure, have a technician perform the test.
6. Record the reading and the conditions under which it was measured. The ATX +12 V rail is normally expected to remain approximately 11.4–12.6 V under specified operating conditions; a reading below 11.5 V can suggest a failing/overloaded PSU, poor connection, or a measurement/setup problem. Further testing under load is needed to confirm the cause.

Example result (illustrative only): 11.3 V DC — below 11.5 V, so the PSU's 12V output may be weak or unstable. This is not an actual measurement; record your own value if you test the unit.

Conclusion: Always disconnect power before inspecting hardware. Use a multimeter only if you know how to handle its probes safely, and never open a PSU. Example measurements in this worksheet are clearly marked as illustrative.